QNN and Error Correction

Neural-Network Decoding of the Seven-Qubit Steane Code

Problem statement: Train a quantum or classical network to decode Steane syndromes, and compare with the lookup table.

1. Problem

Quantum error-correcting codes infer likely physical errors from stabilizer measurement outcomes called syndromes. The seven-qubit Steane code is a CSS code with binary parity checks for bit-flip and phase-flip components. Noisy measurements can make the observed syndrome differ from the syndrome associated with accumulated data errors. This experiment compares a fixed reference lookup decoder with learned classical decoders. The Qiskit QNN was not executable in this runtime, so no QNN score is claimed.

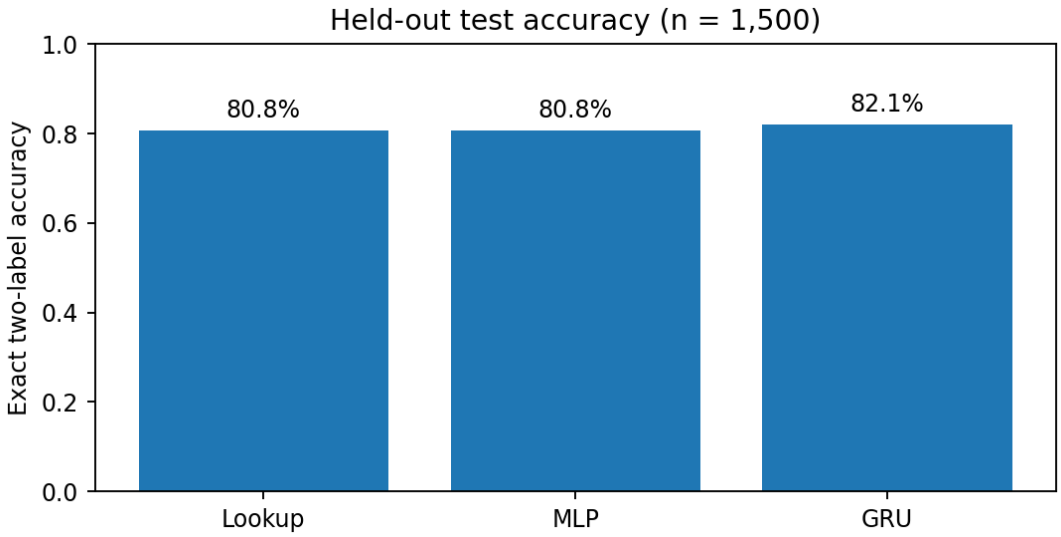
2. Method

The existing saved dataset was reused: 6,000 training, 1,200 validation, and 1,500 test trajectories. Each trajectory has three rounds of six observed bits. Data-error probabilities were p = 0.02, 0.06, 0.12; measurement-bit flip probabilities were q = 0.00, 0.03, 0.08. Fixed split seeds were 2026, 2027, and 2028. Pauli errors accumulate across rounds without intermediate correction.

Syndrome bits are ordered [H x (3 bits), H z (3 bits)] with modulo-2 arithmetic. The target consists of two reference labels (X and Z), each in classes 0–7. The lookup rule and MLP use the latest observed round; the GRU receives all three rounds. The MLP is 6→32→16 with Tanh; the GRU has input size 6 and hidden size 24, followed by 16 logits. Both were trained for 15 epochs with PyTorch, Adam, cross-entropy, and validation-based checkpoint selection.

3. Results

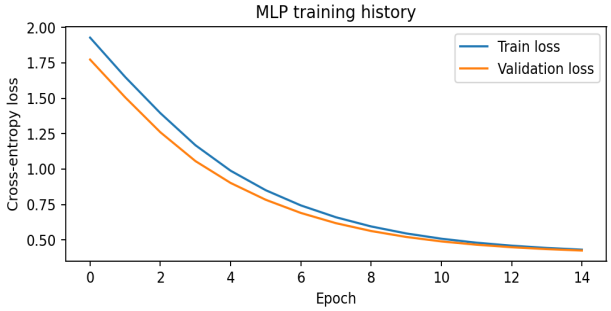
Model	Rounds	Exact	X acc.	Z acc.	Test loss	Train
Lookup	1	80.80%	89.40%	89.87%	N/A	N/A
MLP	1	80.80%	89.40%	89.87%	0.511	1.62s
GRU	3	82.07%	89.87%	90.60%	0.396	2.63s
QNN	—	Not run	Not run	Not run	Not run	Not run

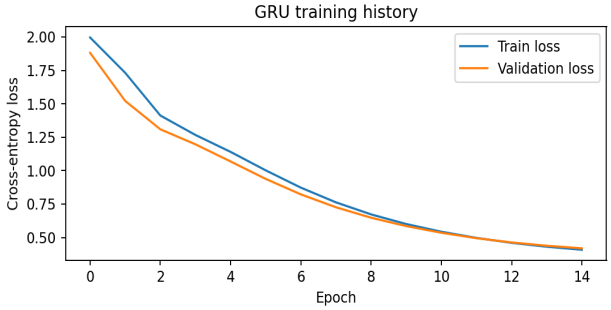


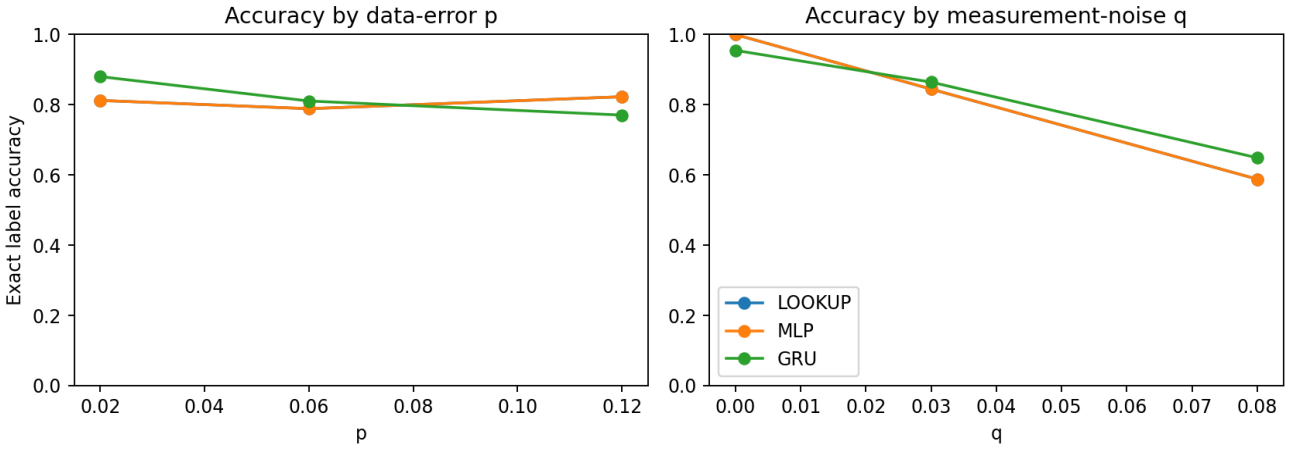
The GRU reached 82.07% exact label accuracy, versus 80.80% for lookup and 80.80% for MLP. The GRU's gain over lookup was 1.27 percentage points, but the GRU receives three rounds while the other two use one. The MLP matched lookup on exact accuracy.

Results: training and noise sensitivity

The training curves below show loss decreasing during the completed classical runs. The noise breakdown reports measured test accuracy for each configured p and q value; the number of examples per subgroup varies because p and q were sampled independently for each trajectory.







Measured single-sample inference latency was 0.019 ms for MLP and 0.151 ms for GRU in this CPU runtime (100 forward passes after 10 warm-ups). These values are host-dependent and are not universal benchmarks.

4. Honest Limits

• The lookup mapping is a single-qubit reference rule, not an optimal general decoder for multi-qubit errors; different physical errors can share a syndrome.

• Reference-label accuracy does not demonstrate successful physical recovery or logical-error suppression. A logical-error rate was not measured because complete encoded-state recovery and logical-observable evaluation were not implemented.

• MLP and lookup use one round, while GRU uses three, so this is not a strictly equal-input comparison.

• The QNN was not run: Qiskit packages were missing, and installation failed because the runtime could not reach the package index. No quantum hardware was used.

• The experiment uses synthetic data and a limited 15-epoch training budget. The scores depend on the specified noise model and saved test split.

Conclusion

Among the completed experiments, the three-round GRU had the highest test exact-label accuracy, while the MLP matched the lookup baseline. The QNN comparison remains incomplete and must be run in a runtime with compatible Qiskit dependencies. The results do not establish quantum advantage.